Chapter4: **Modern Physics**

Lesson1: **Solid-State Physics**

The rapid development of solid-state physics has driven remarkable advances in modern technology. Today, electronic devices such as radios, televisions, LCD or LED TVs, DVD/VCD players, cameras, computers, and mobile phones play an essential role in everyday life.

1. Semiconductor

A semiconductor is a material whose electrical conductivity lies between that of a conductor and an insulator. At room temperature, it does not conduct electricity as easily as a conductor. However, as the temperature increases, its electrical conductivity increases because more charge carriers become available to conduct electric current.

2. Electrical conductivity of semiconductor

The **electrical conductivity** of a semiconductor depends on the movement of **free electrons** and **holes** within its crystal structure. These charge carriers allow electric current to flow through the semiconductor. An important property of a semiconductor is its electrical resistivity, which determines how strongly it opposes the flow of electric current. Materials with **high electrical conductivity** have **low resistivity**, whereas materials with **low electrical conductivity** have **high resistivity**. Therefore, **resistivity** is commonly used to compare the electrical properties of different materials.

The resistance of a conductor depends on its resistivity, length, and cross-sectional area, and is given by

$$R = \rho \frac{l}{A}$$

or equivalently,

$$\rho = \frac{RA}{l}$$

where

- R is resistance (Ω)
- ρ is resistivity $(\Omega \cdot \mathrm{m})$
- l length of the conductor (m)
- A cross-sectional area of the conductor (m^2)

The table below compares the electrical properties of conductors, semiconductors, and insulators using their electrical resistivity. Resistivity is used to indicate how well a material conducts electric current. Materials with low resistivity have high electrical conductivity, whereas materials with high resistivity have low electrical conductivity.

Conductor	Semiconductor	Insulator
Copper (Cu): $\rho = 1.72 \times 10^{-8}\ \Omega\text{m}$	Germanium (Ge): $\rho = 4.6 \times 10^{-1}\ \Omega$	Glass: $\rho = 10^{12}\ \Omega\text{m}$
Silver (Ag): $\rho = 1.60 \times 10^{-8}\ \Omega\text{m}$	Silicon (Si): $\rho = 6.4 \times 10^{2}\ \Omega\text{m}$	Solid rubber (elastic): $\rho = 10^{13}\ \Omega\text{m}$
Gold (Au): $\rho = 2.4 \times 10^{-8}\ \Omega\text{m}$		Mica: $\rho = 10^{15}\ \Omega\text{m}$

Example: A wire has a cross-sectional area of $A = 5.06 \times 10^{-7}\ \text{m}^2$, a length of $l = 20\ \text{m}$, and a resistance of $R = 0.68\ \Omega$. Calculate the **resistivity of the wire** and identify the **type of material**.

Solution: Calculate the **resistivity of the wire** and identify the **type of material**.

Formula: $R = \rho \dfrac{l}{A}$

Since: $A = 5.06 \times 10^{-7}\ \text{m}^2$, $l = 20\ \text{m}$, $R = 0.68\ \Omega$

This give: $R = (5.06 \times 10^{-7}) \dfrac{20}{0.68} = 1.72 \times 10^{-8}\ \Omega.\text{m}$

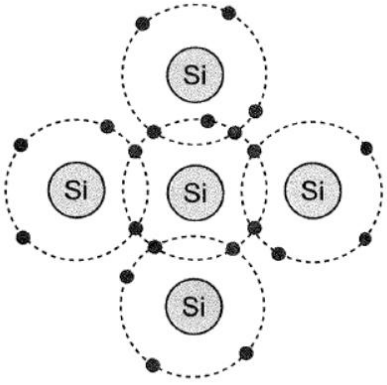
Hence: $R = 1.72 \times 10^{-8}\ \Omega.\text{m}$

By comparing this value with standard resistivity values, the wire is identified as **copper**, which is a **good conductor of electricity**.

Semiconductor materials that are commonly used include germanium (Ge) and silicon (Si). Silicon and germanium are important materials in the field of electronics because they are widely used to manufacture electronic components such as diodes, transistors, integrated circuits (ICs), and other semiconductor devices.

Semiconductors have unique electrical properties. Their electrical conductivity can be significantly controlled by adding a small amount of impurity atoms through a process called doping. In addition, the electrical properties of semiconductors can change depending on external factors such as temperature and incident light intensity.

1.2 Semiconductor

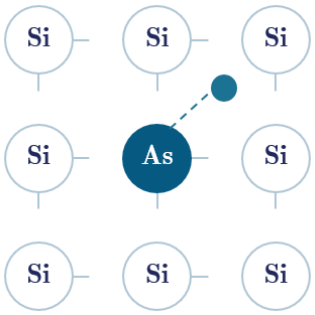


An intrinsic (pure) semiconductor has very low electrical conductivity at room temperature because it contains only a small number of free charge carriers. However, when the temperature increases, more electrons gain enough energy to become free charge carriers, causing the electrical conductivity of the semiconductor to increase. This temperature-dependent behavior is one of the important properties of semiconductors.

By adding a very small amount of impurity atoms (approximately 1 impurity atom per 10 million semiconductor atoms) through a process

called doping, the electrical properties of a pure semiconductor can be modified. This process produces two types of extrinsic semiconductors: N-type semiconductors and P-type semiconductors.

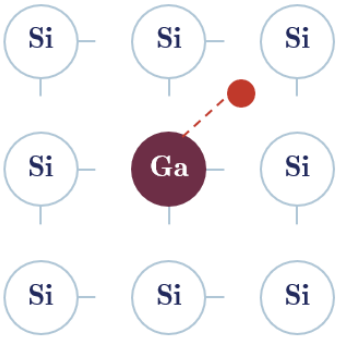
1.2.1 N-Type Semiconductor



An N-type semiconductor is a type of extrinsic semiconductor formed by adding a small amount of impurity atoms that have five valence electrons to a pure silicon semiconductor. These impurity materials include arsenic (As), antimony (Sb), and phosphorus (P). When these impurity atoms combine with silicon atoms, four of their valence electrons form covalent bonds with neighboring silicon atoms, while the fifth valence electron remains free. This free electron becomes the main charge carrier in the N-type semiconductor.

The impurity atoms added to a semiconductor are called donor atoms because they provide extra free electrons. Since the majority charge carriers in this type of semiconductor are negatively charged electrons, it is called an N-type semiconductor, where "N" represents the negative charge of the charge carriers.

1.2.2. P-Type Semiconductor



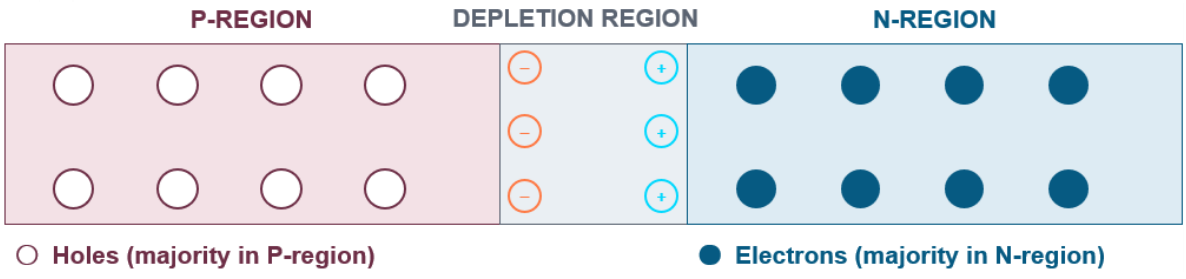
A P-type semiconductor is a type of extrinsic semiconductor formed by adding impurity atoms that have three valence electrons to a pure semiconductor. These impurity atoms are called acceptor atoms. Common materials used to create a P-type semiconductor include boron (B), indium (In), and gallium (Ga).

For example, a boron (B) atom has three valence electrons. When a boron atom is added to silicon, its three valence electrons form covalent bonds with neighboring silicon atoms. However, because boron has only three valence electrons, it cannot complete all four covalent bonds with silicon. This creates a vacancy in the covalent bond, which is called a hole.

A hole behaves like a positive charge because it can attract an electron from a neighboring atom to fill the empty space. The movement of these holes acts as the main charge carrier in a P-type semiconductor.

Therefore, the semiconductor is called a P-type semiconductor, where "P" represents the positive charge of the majority charge carriers.

1.3. Moving of electron and hole



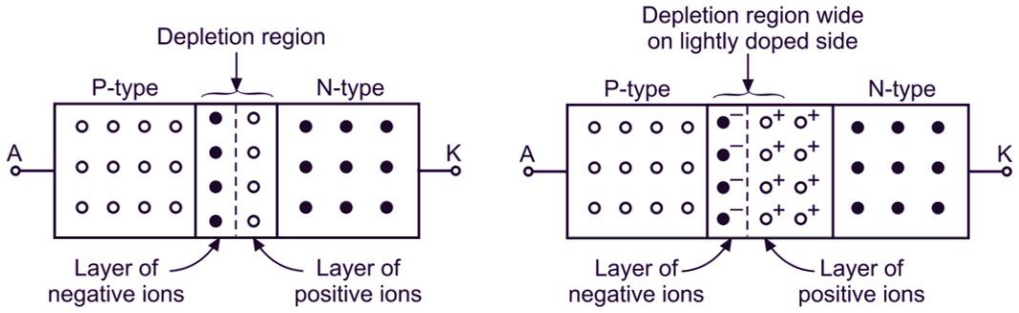
Due to the special properties of atoms, only electrons can move from one atom to another, creating an electric current known as electron current. When an electron moves away from an atom in a crystal lattice, that atom becomes deficient in electrons and forms a positive charge. This missing electron is called a hole, and the magnitude of the hole's charge is equal to the charge of an electron. Therefore, a hole can be considered as a charge carrier that moves in the opposite direction to the movement of electrons.

In an N-type semiconductor, donor atoms such as arsenic (As) have one extra valence electron that does not participate in covalent bonding with silicon atoms. This free electron can move freely through the semiconductor lattice from one atom to another, contributing to electrical conduction. When a vacancy or hole exists, an electron may move to fill that vacancy. Therefore, in an N-type semiconductor, free electrons are the majority charge carriers and are responsible for carrying electric current, while holes are the minority charge carriers.

In a P-type semiconductor, acceptor atoms such as boron (B) form positive ions after accepting electrons from neighboring silicon atoms. These ions remain fixed in the crystal lattice and cannot move. The holes created by the missing electrons become the main charge carriers. Therefore, in a P-type semiconductor, holes are the majority charge carriers, while electrons generated in the semiconductor are the minority charge carriers.

1.4. P-N Junction Layer

Many electronic devices are made by combining N-type and P-type semiconductor crystals, as shown in the figure. The boundary region formed between the N-type and P-type semiconductor materials is called a P-N junction layer.



Due to the attraction between positive and negative charges, electrons from the N-type region move across the P-N junction into the P-type region and recombine with holes near the junction. As electrons leave the N-type region and fill the holes in the P-type region, a region with no free charge carriers is formed. This region is called the depletion region. The depletion region contains fixed ions but has almost no mobile charge carriers because electrons and holes have recombined.

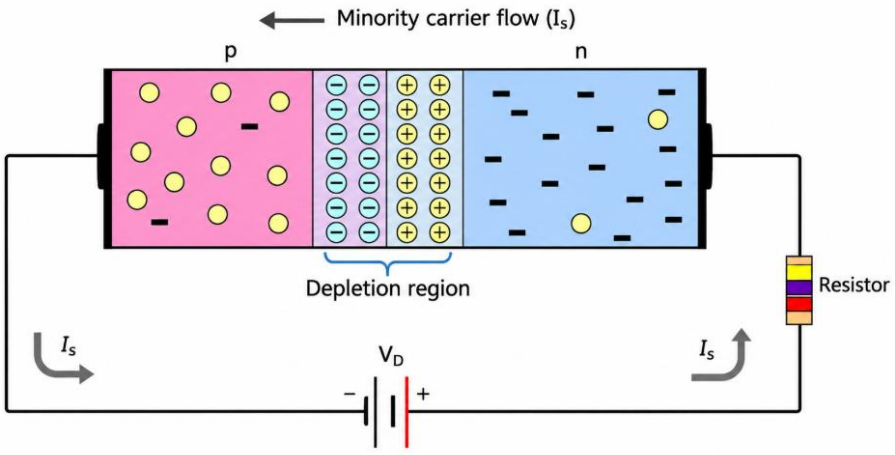
When electrons move from the N-type region across the P-N junction and combine with holes in the P-type region, an electric potential difference is created across the depletion region. This potential

difference produces an electric field that opposes further movement of electrons from the N-type region to the P-type region. The potential barrier is approximately 0.3 V for germanium (Ge) and 0.7 V for silicon (Si).

To control the behavior of the P-N junction, an external voltage is applied across the two terminals of the semiconductor crystal. This process is called biasing, which has two possible conditions:

1.4.1. Forward Bias

When the two terminals of a P-N junction semiconductor are connected to a power supply, with the P-type region connected to the positive $(+)$ terminal and the N-type region connected to the negative $(-)$ terminal, the P-N junction is said to be forward biased. As shown in the figure, the depletion region becomes narrower, allowing electric current to flow more easily through the junction in the conventional direction of current from the P-type region to the N-type region.



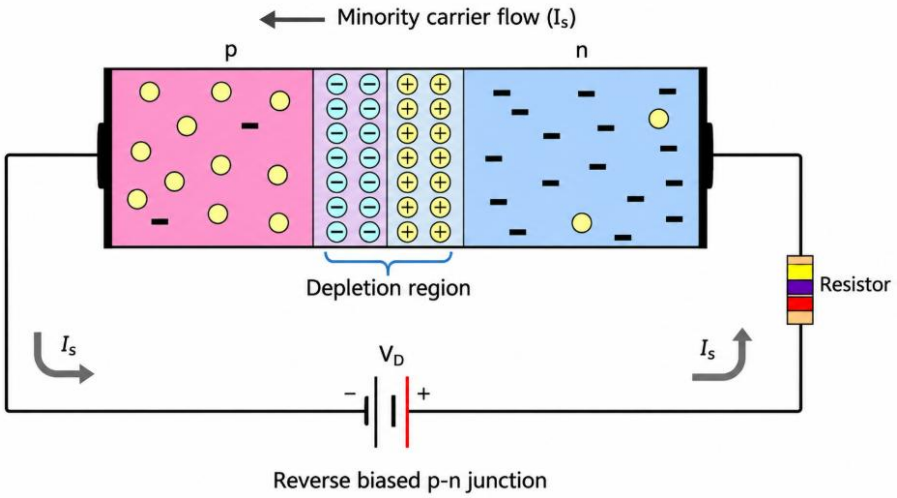
During forward bias, electrons from the N-type region move toward the P-N junction and combine with holes in the P-type region. At the same time, holes from the P-type region move toward the junction and combine with electrons from the N-type region. This movement of charge carriers reduces the width of the depletion region, allowing more charge carriers to cross the junction.

As the applied voltage from the power supply increases, the depletion region becomes narrower. When the applied voltage is increased from 0 V to approximately 0.3 V for germanium (Ge) or 0 V to approximately 0.7 V for silicon (Si), only a small current flows through the semiconductor because the applied voltage is still lower than the potential barrier of the P-N junction.

However, when the applied voltage becomes greater than the potential barrier voltage (approximately 0.3 V for Ge or 0.7 V for Si), the opposing electric field of the depletion region is overcome. As a result, charge carriers can move across the junction easily, causing the electric current to increase significantly.

1.4.2. Reverse Bias

In reverse bias, the polarity of the external voltage source is connected opposite to the forward bias condition. The positive $(+)$ terminal of the power supply is connected to the N-type region, and the negative $(-)$ terminal is connected to the P-type region, as shown in the figure.



In this condition, the positive terminal of the power supply attracts free electrons from the N-type region, causing the electrons near the junction to move away from the P-N junction. At the same time, holes in the P-type region are attracted toward the negative terminal of the power supply. As a result, the depletion region becomes wider, creating a stronger potential barrier that prevents the majority charge carriers from crossing the junction.

Therefore, only a very small reverse current can flow through the P-N junction due to the movement of minority charge carriers. This small current is called the leakage current. In an ideal condition, the current does not flow through the P-N junction during reverse bias; however, in reality, a small leakage current always exists.

Semiconductor devices based on P-N junctions have many applications in modern electronics, including diodes, transistors, integrated circuits (ICs), and other electronic components.

2. Diode

2.1. Definition

A diode is an electronic device made by joining a P-type semiconductor and an N-type semiconductor to form a P-N junction. The semiconductor junction is enclosed in a protective package with two terminals called the anode and the cathode, as shown in the figure.

A diode allows electric current to flow easily in one direction (forward bias) but blocks current flow in the opposite direction (reverse bias). Because of this property, diodes are widely used in electronic circuits for various applications.

Diodes are available in many types; each designed for a specific function in electronic circuits. The common types of diodes include:

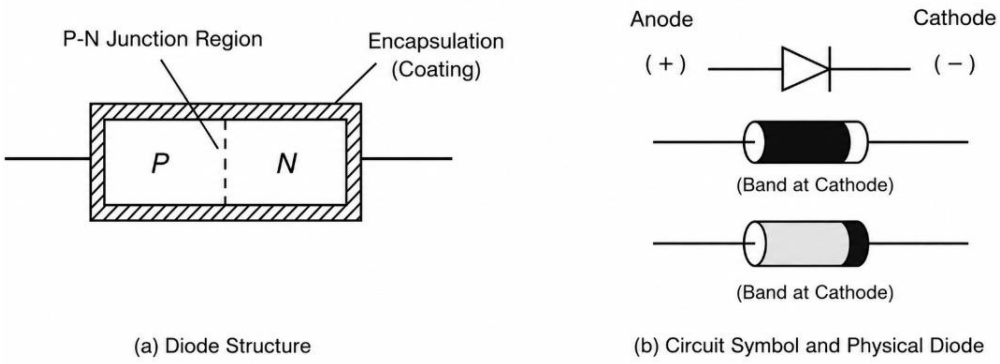
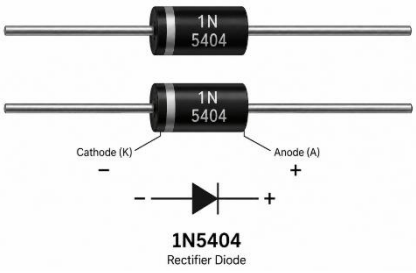


Fig. 4.8

1. Rectifier diode
2. Zener diode
3. Light-emitting diode (LED)
4. Photodiode
5. Optocoupler (Optoisolator) diode
6. Varactor (Varicap) diode
7. Tunnel diode

2.2. Types of Diodes

2.2.1. Rectifier Diode



A rectifier diode, as shown in the figure, is commonly used to convert alternating current (AC) into direct current (DC). It conducts electric current easily in the forward direction but blocks current flow in the reverse direction. During each cycle of an alternating current, a single rectifier diode allows only one half-cycle (either the positive or the negative half-cycle, depending on its orientation) to pass through, producing a half-wave rectified output.

Rectifier diodes are widely used in power supply circuits, battery chargers, and electronic devices that require the conversion of AC to DC. They are also used in signal rectification and other electronic applications.

2.2.2. Zener Diode

A Zener diode is a type of diode that is designed to operate in the reverse-bias region. It is commonly used in electronic circuits to maintain a constant output voltage (regulated voltage). The symbol of a Zener diode is shown in the figure.

When using a Zener diode, the Zener voltage (V_z) should be selected according to the required output voltage. The applied voltage, power dissipation, and operating temperature must not exceed the maximum rated values of the Zener diode to ensure proper operation.

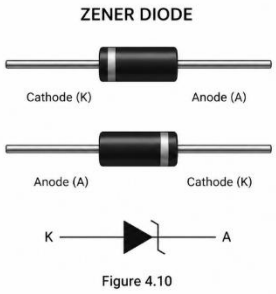


Figure 4.10

The band (or marked end) indicates the cathode (K).
The zener diode symbol has a bent bar at the cathode.



Figure 4.11

Common Zener Diode Series (Examples)	
1N4728A – 1N4764A	0.8 V to 100 V (1 W)
1N5221B – 1N5267B	2.4 V to 75 V (0.5 W)
1N4733A – 1N4756A	5.1 V to 43 V (1 W)

2.2.3. Light-Emitting Diode (LED)

A light-emitting diode (LED) is a semiconductor device that emits light when an electric current passes through it. LEDs are made from special semiconductor materials such as gallium arsenide (GaAs) and gallium phosphide (GaP). They are commonly used as indicator lights in electronic circuits, switches, calculators, digital watches, and many other electronic devices that display numbers or provide visual signals. The symbol of an LED is shown in the figure.

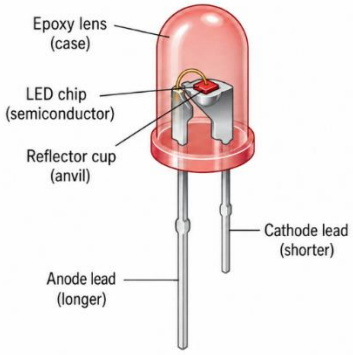


Figure 4.12 Light-Emitting Diode (LED)

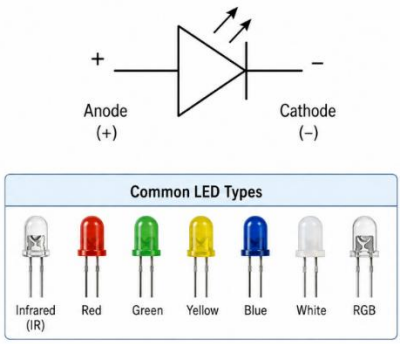


Figure 4.13 LED Symbol and Common Types

2.2.4. Photodiode

A photodiode is a semiconductor device that produces an electric current when it is exposed to light. It is normally operated under reverse-bias conditions, where the incident light generates charge carriers that produce a photocurrent. Photodiodes are commonly used in automatic control circuits, light sensors, optical communication systems, and other light-detecting devices. The symbol of a photodiode is shown in the figure.

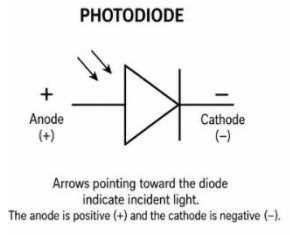




Figure 4.13 Photodiode and Common Packages

2.2.5. Optocoupler (Optoisolator)

An optocoupler is an electronic device that consists of a light-emitting diode (LED) and a photodiode or phototransistor enclosed in the same package. When the LED emits light, the photodiode or phototransistor detects the light and produces an electrical signal. Optocouplers are commonly used in automatic control circuits to provide electrical isolation between two circuits while allowing signals to be transmitted safely. The symbol of an optocoupler is shown in the figure

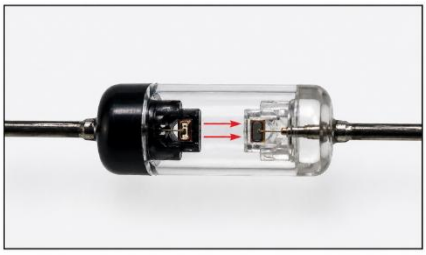


Figure 4.14 Optocoupler (Optoisolator)
An optocoupler consists of an infrared LED and a phototransistor encapsulated in a single package for electrical isolation.

2.2.6. Varactor (Varicap) Diode

A varactor (varicap) diode is a type of diode that operates under reverse-bias conditions. Its capacitance changes with the applied reverse voltage. The symbol of a varactor diode is shown in the figure.

Varactor diodes are commonly used in LC tuning circuits to adjust the resonant frequency and select different communication channels, such as those in radio and television receivers. The application of a varactor diode in an LC circuit is shown in the figure.

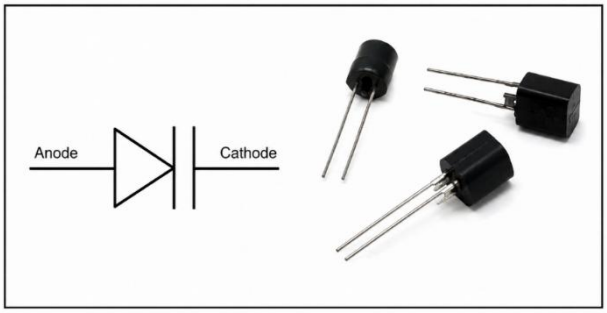


Figure 4.15 Varactor (Varicap) Diode
A varactor (varicap) diode is a reverse-biased PN junction diode whose capacitance varies with the reverse voltage applied.

2.2.7. Tunnel Diode

A tunnel diode is a type of diode that is made by adding a high concentration of impurity atoms to the semiconductor material. Its symbol is shown in the figure. Due to the high level of doping, the P-N junction has a much narrower depletion region than that of an ordinary diode.

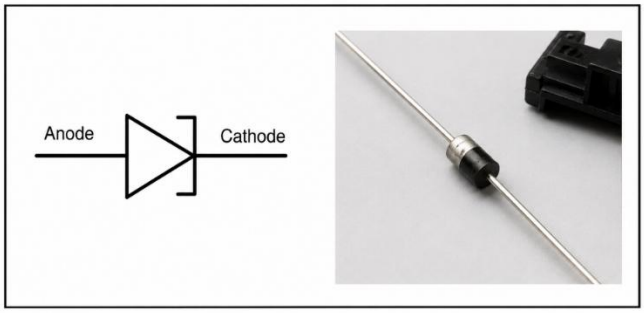


Figure 4.16 Tunnel Diode
A tunnel diode is a heavily doped PN junction diode that operates by quantum mechanical tunneling.

Because of its narrow depletion region, the tunnel diode conducts electric current easily at a low forward voltage (typically below 0.7 V). The current increases rapidly even at a small applied voltage. Due to its fast-switching characteristics, tunnel diodes are commonly used in amplifier circuits, high-speed switching circuits, and oscillator circuits.

3. Transistor

A **transistor** is an electronic device invented in **1947** by three physicists: **William Shockley, John Bardeen, and Walter Brattain**. It is a semiconductor device made by doping a semiconductor crystal to form two P-N junctions, creating three semiconductor regions, as shown in the figure.

A transistor with two **N-type** regions separated by a **P-type** region is called an **NPN transistor**. A transistor with two **P-type** regions separated by an **N-type** region is called a **PNP transistor**.

The middle region of a transistor is called the **base (B)**. The two outer regions are called the **collector (C)** and the **emitter (E)**, as shown in the figure.

NPN and PNP transistors cannot be used interchangeably because the direction of current flow and the biasing conditions are opposite. In an **NPN transistor**, electrons are the majority charge carriers and move from the emitter to the collector. In a **PNP transistor**, holes are the majority charge carriers and move from the emitter to the collector.

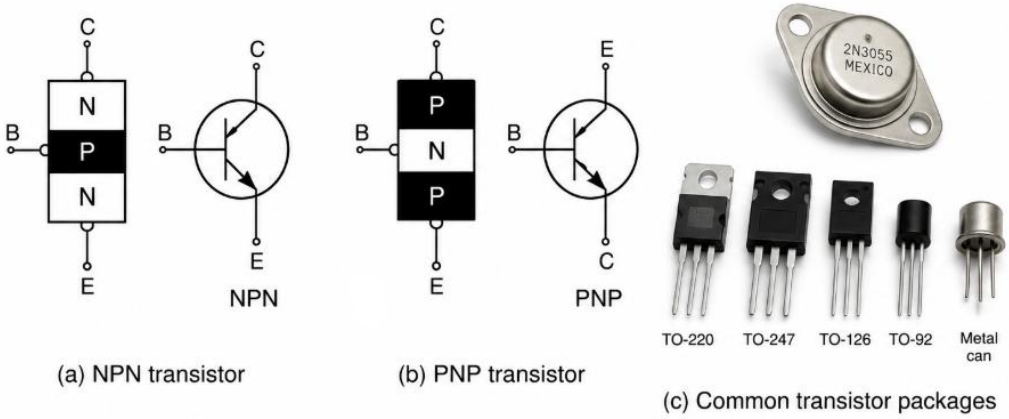


Figure 4.17 Transistor

A transistor is a three-terminal semiconductor device that can be used for amplification and switching.

B = Base C = Collector E = Emitter

To operate properly, each type of transistor must be connected with the correct biasing (polarization) so that electric current can flow through the circuit as required.

3.1. NPN Transistor

An NPN transistor is a type of transistor in which electrons are the majority charge carriers. Electrons move more easily through the semiconductor material than holes because they have higher mobility. Therefore, the collector current is relatively large compared with the base current. During normal operation, the base-emitter junction is forward biased, while the collector-base junction is reverse biased.

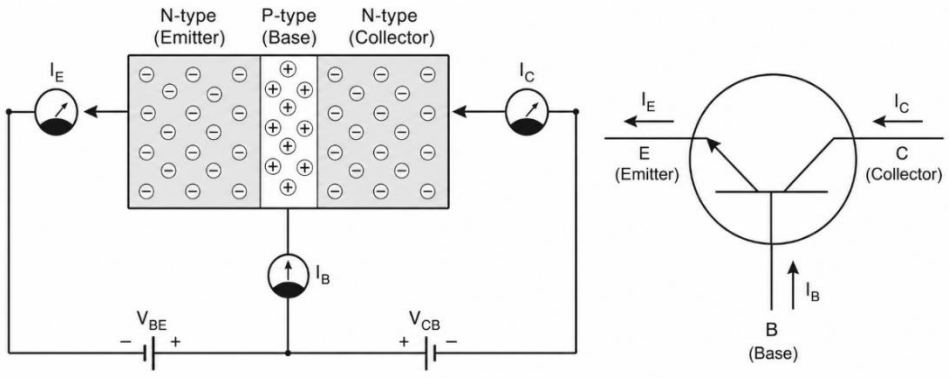


Figure 4.18 *Structure and circuit symbol of an NPN transistor*

The emitter current (I_E) is the largest current in the transistor. Approximately 99% of the charge carriers injected by the emitter reach the collector, forming the collector current (I_C). The remaining 1% recombine in the base region, producing the base current (I_B). Therefore, the relationship between the transistor currents is given by:

$$I_E = I_B + I_C$$

A transistor has the ability to amplify electric current. The current amplification is expressed by the current gain. For a common-emitter transistor, the current gain is denoted by beta (β) and is given by:

$$\beta = \frac{I_C}{I_B}$$

For a common-base transistor, the current gain is denoted by alpha (α) and is given by:

$$\alpha = \frac{I_C}{I_E}$$

Example 1: Calculating the Current Gain (β)

A transistor operating in a DC circuit has a base current of $I_B = 60 \times 10^{-3}\ \text{mA}$ and a collector current of $I_C = 8\ \text{mA}$. Calculate the DC current gain (β).

Solution: Calculate the DC current gain (β).

Formula: $\beta = \frac{I_C}{I_B}$

Since: $I_C = 8\ \text{mA},\ I_B = 60 \times 10^{-3}\ \text{mA}$

This give: $\beta = \frac{8\,\text{mA}}{60 \times 10^{-3}\ \text{mA}}$

Therefore, the DC current gain of the transistor is $\beta = 133.3$

Example 2: Calculating the Current Gain (α)

A transistor operating in the common-base configuration has a base current of $I_B = 120 \times 10^{-3}\ \text{mA}$ and a collector current of $I_C = 16\ \text{mA}$. Calculate the current gain (α).

Solution: Calculate the current gain (α).

Formula: $\alpha = \dfrac{I_C}{I_E}$

Since: $I_E = I_B + I_C = \left(120 \times 10^{-3} + 16\right)\text{mA} = 16.12\,\text{mA}$

This gives: $\alpha = \dfrac{16\,\text{mA}}{16.12\,\text{mA}}$

Therefore, the current gain of the transistor is $\alpha = 0.993$

3.2. PNP Transistor

A PNP transistor is a type of transistor in which the base-emitter junction is forward biased, while the collector-base junction is reverse biased. However, the polarity of the base-emitter voltage (V_{BE}) and the collector-base voltage is opposite to that of an NPN transistor. Therefore, PNP and NPN transistors cannot be used interchangeably because their biasing conditions and the direction of current flow are opposite. In a PNP transistor, the majority charge carriers are holes, and the conventional current flows from the emitter to the collector.

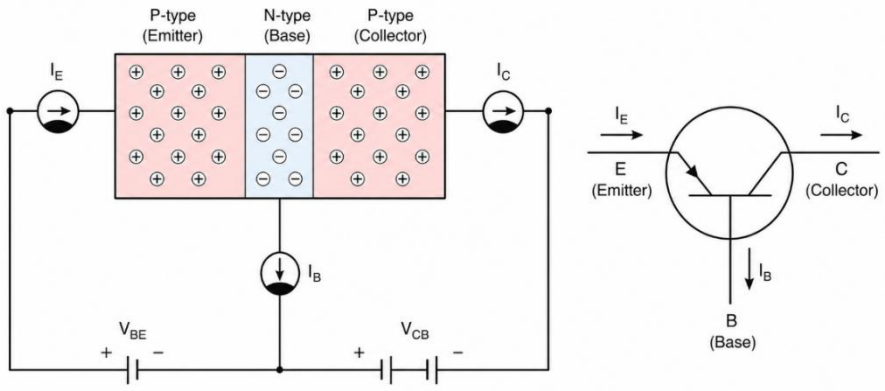


Figure 4.19 *Structure and circuit symbol of a PNP transistor*

Both NPN and PNP transistors have similar applications, but they differ in their biasing conditions and the direction of current flow. In practice, NPN transistors are more commonly used than PNP transistors because electrons have higher mobility than holes and move more easily through the semiconductor material.

Many other types of transistors have been developed for different electronic applications, including:

- **Junction Field-Effect Transistor (JFET):** A JFET controls the flow of current by changing the voltage applied to the gate terminal. The drain current increases or decreases according to the gate voltage. JFETs are classified into two types: **N-channel JFET** and **P-channel JFET**.
- **Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET):** A MOSFET controls the current flowing from the source to the drain by applying a voltage to the gate terminal. It is available in two types: **N-channel MOSFET** and **P-channel MOSFET**.

- **Enhancement-Type MOSFET (E-MOSFET):** An enhancement-type MOSFET differs from a conventional MOSFET because it has no conducting channel between the source and the drain under normal conditions. Therefore, it is a **normally OFF** device. When a suitable voltage is applied to the gate, charge carriers are attracted to form a conducting channel between the source and the drain. As the gate voltage increases, the conductivity of the channel also increases.

In electronics, transistors are widely used in many circuits, such as volume control circuits, sound control circuits, audio amplifiers, voltage regulators, automatic switching circuits, and many other electronic devices.

4. Integrated Circuit (IC)

4.1. Definition

The integrated circuit (IC) was first proposed by the British engineer Geoffrey W. A. Dummer in 1952. The first practical integrated circuits were later developed independently by Jack Kilby in 1958 and Robert Noyce in 1959.

In electronic circuits, transistors, resistors, diodes, and other electronic components are fabricated together on a small silicon chip using photolithography and semiconductor manufacturing processes, as shown in the figure. This device is called an integrated circuit (IC).

An integrated circuit is a complete electronic circuit built on a single silicon chip. It contains many electronic components, such as resistors, diodes, transistors, and other semiconductor devices, connected together to perform a specific function. The IC is enclosed in a protective package with external terminals (pins) for connection to an electronic circuit.

Integrated circuits are manufactured in many different package types. Some ICs have three pins similar to a transistor, while others have pins arranged in a single row, two parallel rows, or around all four sides of the package, depending on their design and application.

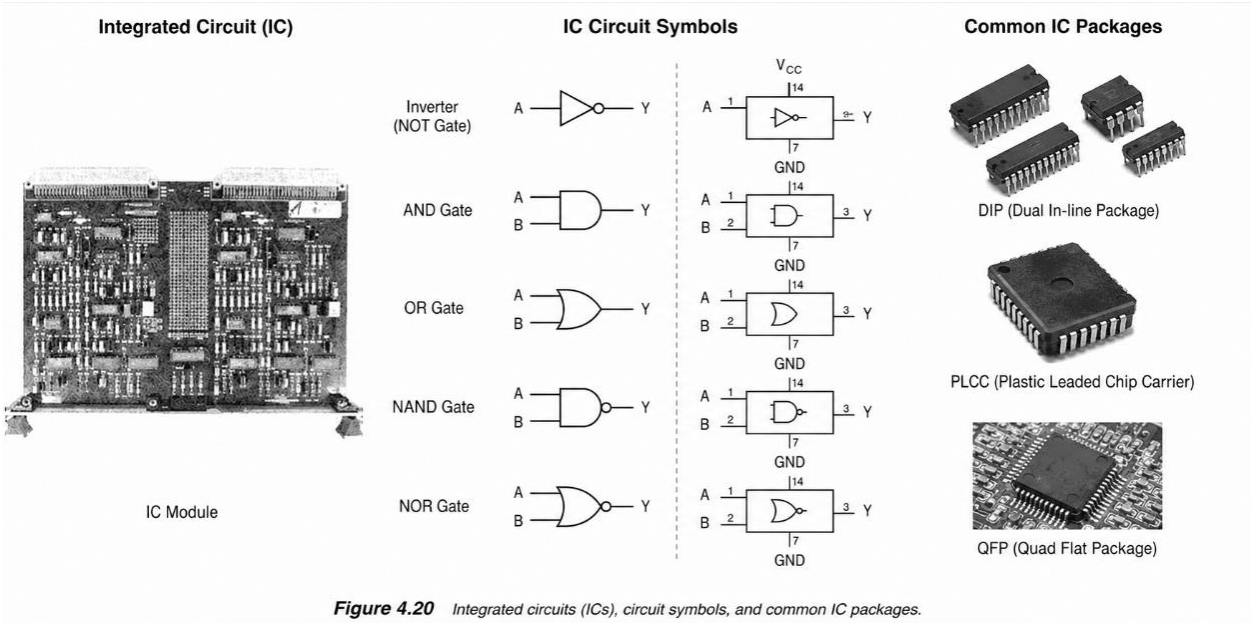


Figure 4.20 *Integrated circuits (ICs), circuit symbols, and common IC packages.*

Lesson2: **Special Theory of Relativity**

1. Galilean Coordinate Transformation

1.1. Reference Frame

When studying the motion of an object, it is necessary to determine a suitable reference frame. A reference frame is a coordinate system used to describe the position and motion of an object.

A reference frame is called an inertial frame if Newton's laws of motion are valid in that frame. Any reference frame that moves at a constant velocity relative to an inertial frame is also considered an inertial frame.

In classical mechanics, a reference frame attached to the Earth's surface is often assumed to be an inertial frame because the effects of Earth's rotation and revolution are very small for many everyday situations.

The Earth has two main motions:

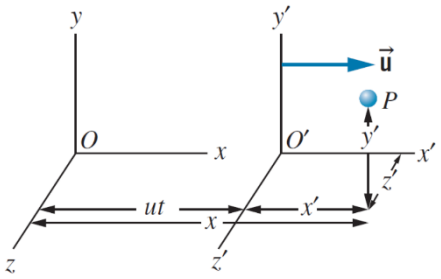
- Orbital motion around the Sun with a speed of approximately 3×10^4 m/s.
- Rotational motion around its axis with a speed of approximately $460\ \mathrm{m/s}$ at the equator.

The reason that the Earth can be considered an approximately inertial reference frame is that its rotational speed is much smaller than its orbital speed, and the changes in velocity over a short period of time are negligible for ordinary motion

2. Galilean Coordinate Transformation

2.1. Galilean Coordinate Transformation Equation

Consider two observers located in two inertial reference frames, S and S'. The reference frame S has coordinates $S\ (x,\ y,\ z,\ t)$, while the reference frame S' has coordinates $S'\ (x',\ y',\ z',\ t')$, The frame S' moves along the positive x-axis with a constant velocity u relative to frame S, as shown in the figure.



We assume that time is the same in both reference frames:

$$t = t'$$

At the instant $t = 0$, the origins of the two reference frames, O and O', coincide with each other. If an object is located at point P, the Galilean coordinate transformation equations are given as:

$$x' = x - ut, y' = y, z' = z$$

To find the velocity of the object as observed from frames S and S', we take the derivative of these equations with respect to time. Since we assume $t = t'$, the Galilean velocity transformation equations are:

$$\begin{aligned}\frac{dx'}{dt'} &= \frac{dx}{dt} - u\frac{dt}{dt} = v_x - u \\ \frac{dy'}{dt'} &= \frac{dy}{dt} = v_y \\ \frac{dz'}{dt'} &= \frac{dz}{dt} = v_z\end{aligned} \quad \text{or} \quad \begin{cases} v_x' = v_x - u \\ v_y' = v_y \\ v_z' = v_z \end{cases}$$

where:

- v_x is the velocity component of the object along the x-axis measured in frame S.
- u is the constant velocity of frame S' relative to frame S.
- v_x' is the velocity component of the object along the x-axis measured in frame S'.

Similarly, taking the derivative of the velocity equations with respect to time gives the relationship between the accelerations:

$$\begin{cases} a_x' = a_x \\ a_y' = a_y \\ a_z' = a_z \end{cases}$$

where:

- a is the acceleration of the object measured in frame S.
- a' is the acceleration of the object measured in frame S'.
- u is the velocity of frame S' relative to frame S.
- v and v' are the velocities of the object measured from frames S and S', respectively.

Example: Two cars, A and B, are traveling at constant speeds along a straight road in the same direction. Car A moves at $60\ \mathrm{km/h}$ and car B moves at $40\ \mathrm{km/h}$, where both speeds are measured relative to an observer on the ground, as shown in **Figure 2.3a**. Determine the speed of car A relative to car B.

Solution: Determine the speed of car A relative to car B.

We choose the observer on the ground as reference frame S, and car B as reference frame S'. The frame S' moves at a constant velocity $u = v_B = 40\ \mathrm{km/h}$ relative to frame S. Car A moves with velocity $v_x = v_A = 60\ \mathrm{km/h}$ relative to frame S.

Using the Galilean velocity transformation equation:

$$v_x' = v_x - u$$

Substitute the values:

$$v_x' = (60 - 40)\,\mathrm{km/h} = 20\,\mathrm{km/h}$$

Therefore, the speed of car A relative to car B is $20\ \mathrm{km/h}$

2. Speed of Light and the Michelson–Morley Experiment

• The purpose of the Michelson–Morley experiment was to measure the speed of the Earth relative to the hypothetical luminiferous ether (vacuum), which was assumed at that time to be an absolute stationary reference frame.

• The instrument used in this experiment is called the Michelson interferometer. An interferometer is an optical instrument used to measure the difference in the path lengths of two light beams by observing their interference pattern.

• A monochromatic light beam is directed toward a beam splitter M_0, where it is divided into two beams. One beam travels toward plane mirror M_1, while the other beam travels toward plane mirror M_2.

• The two light beams are reflected back from mirrors M_1 and M_2. After returning to the beam splitter M_0, they recombine and travel toward the telescope, where the interference pattern is observed.

• Assume that the Earth moves through the ether with a speed v. If the ether exists, the speed of light measured along the direction of the Earth's motion would differ from the speed measured perpendicular to the Earth's motion.

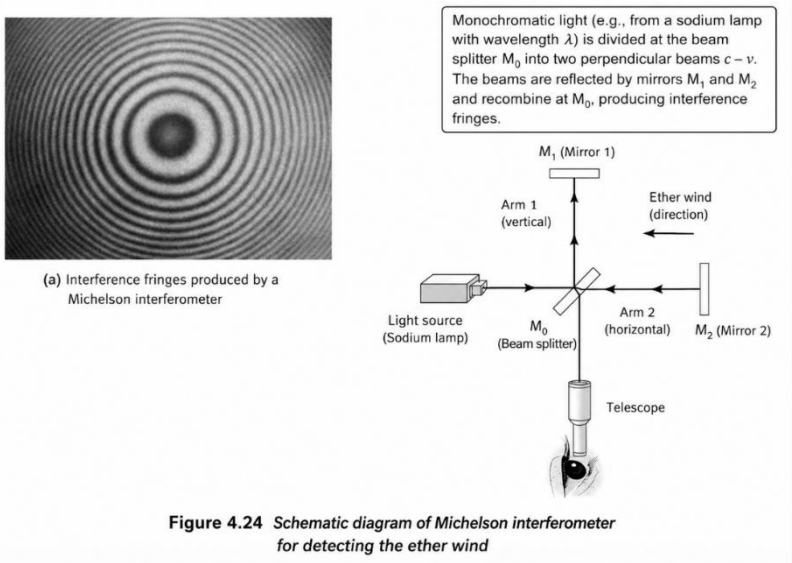


Figure 4.24 *Schematic diagram of Michelson interferometer for detecting the ether wind*

2.1. Explanation of the Michelson–Morley Experiment

Assume that the distances M_0M_1 and M_0M_2 are equal, such that

$$M_0M_1 = M_0M_2 = L$$

According to the Galilean transformation, if the Earth moves through the hypothetical ether with speed v, the speed of light along the direction of the Earth's motion is c-v in one direction and $c + v$ in the opposite direction.

The time taken for the first light beam to travel from M_0 to M_1 is

$$t_{\text{forward}} = \frac{L}{c - v}$$

The time taken for the reflected beam to return from M_1 to M_0 is

$$t_{\text{backward}} = \frac{L}{c + v}$$

Therefore, the total travel time of the first beam is

$$t_1 = \frac{L}{c-v} + \frac{L}{c+v} = \frac{2Lc}{c^2 - v^2} = \frac{2L}{c}\left(1 - \frac{v^2}{c^2}\right)^{-1}$$

For the second light beam travelling perpendicular to the Earth's motion, the effective speed is obtained using the Pythagorean theorem. Therefore, the total travel time is

$$t_2 = \frac{2L}{\sqrt{c^2 - v^2}}$$

Or

$$t_2 = \frac{2L}{c}\left(1 - \frac{v^2}{c^2}\right)^{-1/2}$$

The difference in travel time between the two light beams is

$$\Delta t = t_1 - t_2$$

$$\Delta t = \frac{2L}{c}\left[\left(1 - \frac{v^2}{c^2}\right)^{-1} - \left(1 - \frac{v^2}{c^2}\right)^{-1/2}\right]$$

Since $\frac{v}{c} \ll 1$

the Binomial Expansion can be used:

$$(1-x)^n \approx 1 - nx, x \ll 1$$

Hence $\left(1 - \frac{v^2}{c^2}\right)^{-1} \approx 1 + \frac{v^2}{c^2}$

And $\left(1 - \frac{v^2}{c^2}\right)^{-1/2} \approx 1 + \frac{v^2}{2c^2}$

Therefore $\Delta t = \frac{2L}{c}\left[1 + \frac{v^2}{c^2} - 1 - \frac{v^2}{2c^2}\right] = \frac{Lv^2}{c^3}$

The corresponding path difference is

$$\Delta x = c\Delta t$$

Therefore, $\Delta x = \frac{Lv^2}{c^2}$

The expected fringe shift is

$$\Delta N = \frac{\Delta x}{\lambda}$$

Hence, $$\Delta N = \frac{Lv^2}{\lambda c^2}$$

Numerical Calculation

Given: $$L = 11\ \text{m}, v = 3.0 \times 10^4\ \text{m/s}, \lambda = 500\,\text{nm} = 500 \times 10^{-9}\ \text{m}$$

Substituting these values,

$$\Delta N = \frac{11(3.0 \times 10^4)^2}{(500 \times 10^{-9})(3.0 \times 10^8)^2}$$

Hence: $$\Delta N = 0.22$$

Conclusion

Michelson and Morley expected to observe a fringe shift of approximately 0.22 fringe. Since their interferometer could detect fringe shifts as small as 0.01 fringe, this shift should have been measurable. However, no fringe shift was observed when the apparatus was rotated. This result is known as the *null result* of the Michelson–Morley experiment.

From this experiment, they concluded that there was no evidence for the existence of the luminiferous ether (the absolute stationary reference frame). The null result showed that the speed of light is the same in all directions, regardless of the motion of the Earth. This discovery became one of the experimental foundations of Einstein's Special Theory of Relativity.

3. Einstein's Postulates

The speed of light in a vacuum is represented by the symbol c, and its value is approximately:

$$c = 3 \times 10^8\ \text{m/s}$$

The accurate measured value of the speed of light is:

$$c = 299792458\ \text{m/s}$$

According to Maxwell's electromagnetic theory, the speed of light can be expressed as:

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}$$

where:

- ε_0 is the permittivity of free space (vacuum permittivity).
- μ_0 is the permeability of free space (vacuum permeability).

The Special Theory of Relativity is based on two postulates proposed by Albert Einstein in 1905:

- **The Principle of Relativity:** The laws of physics are the same in all inertial reference frames.
- **The Principle of the Constancy of the Speed of Light:** The speed of light in free space has the same value c in all inertial reference frames.

The first postulate states that the laws of physics are absolute, universal, and identical for all inertial observers. A physical law that is valid for one inertial observer must also be valid for every other inertial observer.

The second postulate states that light always travels through a vacuum at a constant speed c, regardless of the motion of the light source or the observer. The speed of light does not depend on the velocity of the object emitting the light or the observer measuring it.
The Special Theory of Relativity applies only to inertial reference frames, which are reference frames moving at constant velocity relative to each other without acceleration.

4. Lorentz Transformation

We have studied that the Galilean transformation describes the relationship between coordinates in two inertial reference frames, where frame S' moves at a constant velocity u relative to frame S. In Galilean transformation, we assume that time is the same in both reference frames:

$$t = t'$$

However, the Galilean transformation is only valid when the relative speed between the two reference frames is much smaller than the speed of light:

$$u << c$$

When the velocity $u$approaches the speed of light, the Galilean transformation is no longer accurate. Therefore, in this section, we study another coordinate transformation called the Lorentz transformation. The Lorentz transformation is more general than the Galilean transformation because it can be applied for all velocities between zero and the speed of light:

$$0 < u < c$$

An important difference between the Galilean transformation and the Lorentz transformation is that the Lorentz transformation does not assume that time is the same in all inertial reference frames. The time interval measured in one inertial reference frame can be different from the time interval measured in another inertial reference frame.

1. Lorentz Coordinate Transformation

For the condition shown in Figure 1 (used in the Galilean transformation), the Lorentz coordinate transformation equations can be written as follows:

Transformation from S to S'

$$x' = \gamma(x - ut), y' = y, z' = z, t' = \gamma\left(t - \frac{ux}{c^2}\right)$$

Reverse transformation from S' to S

$$x = \gamma(x' + ut), y = y', z = z', t = \gamma\left(t' + \frac{ux'}{c^2}\right)$$

where the Lorentz factor γ is defined as:

$$\gamma = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}}$$

The properties of the Lorentz factor are:

- $\gamma > 1$ when u approaches the speed of light $(u \to c)$.
- $\gamma \approx 1$ when u is much smaller than the speed of light $(u << c)$.

2. Equation of Lorentz Transformation

The velocity components of an object observed in each reference frame are denoted as follows:
For frame S:

$$v_x = \frac{dx}{dt}, v_y = \frac{dy}{dt}, v_z = \frac{dz}{dt}$$

For frame S':

$$v'_x = \frac{dx'}{dt'}, v'_y = \frac{dy'}{dt'}, v'_z = \frac{dz'}{dt'}$$

To obtain the velocity transformation equations, we differentiate the Lorentz coordinate transformation equations with respect to time as follows:

$$dx' = \gamma(dx - udt), dy' = dy, dz' = dz, dt' = \gamma\left(dt - \frac{u}{c^2}dx\right)$$

For the x-direction velocity:

$$v'_x = \frac{dx'}{dt'} = \frac{\gamma(dx - udt)}{\gamma\left(dt - \frac{u}{c^2}dx\right)}$$

Divide the numerator and denominator by dt:

$$v'_x = \frac{\frac{dx}{dt} - u}{1 - \frac{u}{c^2}\frac{dx}{dt}}$$

Therefore:

$$v'_x = \frac{v_x - u}{1 - \frac{u}{c^2}v_x}$$

For the y-direction velocity:

$$v'_y = \frac{dy'}{dt'} = \frac{dy}{\gamma\left(dt - \frac{u}{c^2}dx\right)}$$

Dividing by dt:

$$v'_y = \frac{v_y}{\gamma\left(1 - \frac{u}{c^2}v_x\right)}$$

For the z-direction velocity:

$$v'_z = \frac{dz'}{dt'} = \frac{v_z}{\gamma\left(1 - \frac{u}{c^2}v_x\right)}$$

The Lorentz Velocity Transformation Equations

Transforming from frame S to frame S':

$$v_x' = \frac{v_x - u}{1 - \frac{u}{c^2} v_x}, \; v_y' = \frac{v_y}{\gamma\left(1 - \frac{u}{c^2} v_x\right)}, \; v_z' = \frac{v_z}{\gamma\left(1 - \frac{u}{c^2} v_x\right)}$$

The reverse transformation from frame S' to frame S:

$$v_x = \frac{v_x' + u}{1 + \frac{u}{c^2} v_x'}, \; v_y = \frac{v_y'}{\gamma\left(1 + \frac{u}{c^2} v_x'\right)}, \; v_z = \frac{v_z'}{\gamma\left(1 + \frac{u}{c^2} v_x'\right)}$$

Example1: A spaceship moving away from the Earth at a speed of $0.80c$ fires a missile parallel to its direction of motion (Figure 2.15). The missile moves at a speed of $0.60c$ relative to the ship. What is the speed of the missile as measured by an observer on the Earth?

Solution:

Determine the speed of the missile relative to Earth.

We choose the Earth as the reference frame S, and the spaceship as the reference frame S'. The spaceship moves at a velocity $u = 0.80c$ relative to Earth. The missile moves with a velocity $v_x' = 0.60\,c$ relative to the spaceship S'.

Because the velocity is close to the speed of light, we use the Lorentz velocity transformation equation:

$$v_x = \frac{v_x' + u}{1 + \frac{u}{c^2} v_x'}$$

Substitute the values:

$$v_x = \frac{0.60\,c + 0.80\,c}{1 + \frac{(0.80\,c)(0.60\,c)}{c^2}} = \frac{1.40\,c}{1 + 0.48} = \frac{1.40\,c}{1.48}$$

Therefore, the speed of the missile measured by an observer on Earth is $v_x = 0.95$ c

Example2: A spaceship moving away from the Earth at a speed of $0.50c$ fires a missile parallel to its direction of motion (Figure 2.15). The missile moves at a speed of $0.70c$ relative to the ship. What is the speed of the missile as measured by an observer on the Earth?

Solution:

Determine the speed of the missile relative to Earth.

We choose the Earth as the reference frame S, and the spaceship as the reference frame S'. The spaceship moves with velocity $u = 0.50$ c relative to Earth. The missile moves with velocity $v_x' = 0.70\,c$ relative to the spaceship S'.

Since the velocity of the missile is close to the speed of light, we use the Lorentz velocity transformation equation:

$$v_x = \frac{v_x' + u}{1 + \frac{u}{c^2} v_x'}$$

Substitute the values:

$$v_x = \frac{0.70\,c + 0.50\,c}{1 + \frac{(0.50\,c)(0.70\,c)}{c^2}} = \frac{1.20\,c}{1 + 0.35} = \frac{1.20\,c}{1.35}$$

Therefore, the speed of the missile measured by an observer on Earth is $v_x = 0.99$ c

Example3: Two electrons are moving in opposite directions, each with a speed of $0.50c$ relative to a laboratory reference frame. Determine the speed of the first electron relative to the second electron.

Solution:

Determine the speed of the first electron relative to the second electron.

We choose the laboratory as reference frame S. The second electron is considered as reference frame S', which moves with velocity $u = -0.50$ c relative to the laboratory frame S. The first electron moves with velocity $v_x = 0.50$ c relative to the laboratory frame S.

Because the velocity is close to the speed of light, we use the Lorentz velocity transformation equation:

$$v_x' = \frac{v_x - u}{1 - \frac{u}{c^2} v_x}$$

Substitute the values:

$$v_x' = \frac{0.50\,c - (-0.50\,c)}{1 - \frac{(-0.50\,c)(0.50\,c)}{c^2}} = \frac{1.00\,c}{1 + 0.25} = \frac{1.00\,c}{1.25}$$

Therefore, the speed of the first electron relative to the second electron is $v_x' = 0.80c$

Example4: Two spacecraft A and B are moving in opposite directions as shown in **Figure 39.14**. An observer on the Earth measures the speed of spacecraft A to be $0.750c$ and the speed of spacecraft B to be $0.850c$. Find the velocity of spacecraft B as observed by the crew on spacecraft A.

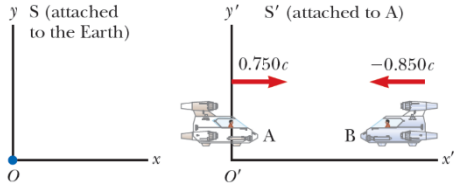


Figure 39.14 (Example 39.6) Two spacecraft A and B move in opposite directions. The speed of spacecraft B relative to spacecraft A is *less* than c and is obtained from the relativistic velocity transformation equation.

Solution:

Find the velocity of spacecraft B as observed by the crew on spacecraft A.

We choose the Earth as the reference frame (S), and spacecraft A as the reference frame (S'). Spacecraft A moves with a velocity $(u = +0.750\text{c})$ relative to Earth. Spacecraft B moves in the opposite direction with a velocity $(v_x = -0.850\text{c})$ relative to Earth.

Because the velocity is close to the speed of light, we use the Lorentz velocity transformation equation:

$$v_x' = \frac{v_x - u}{1 - \frac{u}{c^2} v_x}$$

Substitute the values:

$$v_x' = \frac{-0.85\,c - 0.75\,c}{1 - \frac{(0.75\,c)(-0.85\,c)}{c^2}} = \frac{-1.600\,c}{1 + 0.6375} = \frac{-1.600\,c}{1.6375}$$

Therefore, the velocity of spacecraft B as observed by the crew on spacecraft A is $v_x' = -0.977c$

Lesson3: **Atomic Structure**

1. Introduction

Before the year 1900, the atom was considered to be the smallest fundamental particle of matter. It was believed to have no internal structure and could not be divided into smaller particles. However, some scientists suggested that the atom might contain smaller particles moving inside it.

This idea changed after J. J. Thomson discovered the electron in 1897. Thomson proposed that the electron is a fundamental particle of the atom and can be separated from it. His discovery proved that the atom is not indivisible but has an internal structure.

1. Electron

1.1. Cathode Rays

During the nineteenth century, physicists developed two important devices: the vacuum pump and the induction coil. The vacuum pump was used to produce a very low pressure inside a glass tube, while the induction coil was used to generate a high potential difference between two electrodes in the tube. The vacuum tube provided a suitable environment for accelerating charged particles.

In Thomson's cathode ray tube, a high potential difference is applied between two metal electrodes placed inside the vacuum tube. As a result, the phosphor-coated screen at one end of the tube emits light. Thomson proposed that the light produced on the phosphor-coated screen was caused by a stream of charged particles striking the screen. This stream of charged particles is called a *cathode ray* because it originates from the cathode (the negatively charged electrode). The cathode rays travel at very high speed through the vacuum tube and eventually strike the phosphor-coated screen.

Thomson demonstrated that the cathode ray could be deflected by both a magnetic field and an electric field. The experiment showed that the cathode ray was deflected toward the positive plate, indicating that the particles in the ray carry a negative electric charge.

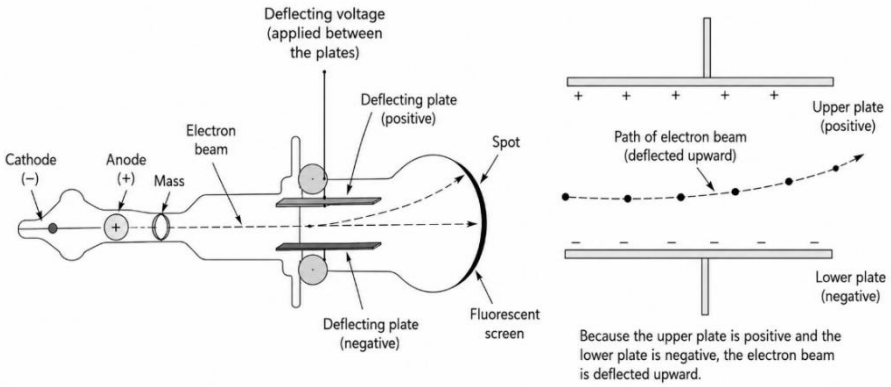


Figure 4.32 *Deflection of an electron beam in a cathode ray tube (CRT)*

From these experiments, Thomson was able to determine the nature of cathode rays. He concluded that cathode rays consist of negatively charged particles called *electrons* (formerly called *negatons*). Electrons have both mass and electric charge. Every electron carries the same negative charge, $-e$, and has a definite mass, m.

1.2. Determination of the Charge-to-Mass Ratio (e/m)

J. J. Thomson measured the ratio of the electric charge to the mass of the electron using cathode rays. This ratio is called the charge-to-mass ratio, denoted by e/m, where e is the electric charge of the electron and m is its mass.

Suppose a charged particle of mass $m$and charge q moves with an initial velocity v_0 perpendicular to a uniform magnetic field B. As studied in Chapter 3, the magnetic force causes the particle to follow a curved path. The relationship is given by:

$$Z = \frac{Dl|q|B}{mv_0}$$

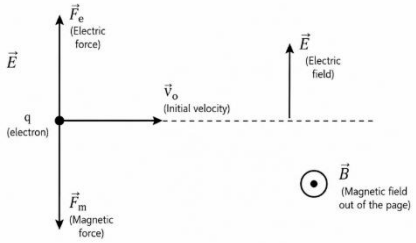


Figure 4.33 *Forces acting on an electron moving to the right in crossed electric and magnetic fields*

Rearranging the equation gives:

$$\frac{q}{m} = \frac{Zv_0}{DlB}$$

Thomson designed an apparatus that allowed charged particles to move with a constant velocity. He applied a uniform electric field E and a uniform magnetic field B, which were perpendicular to each other. When the electric force and the magnetic force were equal in magnitude and opposite in direction, the charged particles passed through the fields in a straight line without deflection.

Since the electric force equals the magnetic force:

$$F_E = F_B$$

$$qE = qv_0B$$

Therefore,

$$v_0 = \frac{E}{B}$$

Substituting this expression into the equation for q/m gives:

$$\frac{q}{m} = \frac{Z}{DlB}\left(\frac{E}{B}\right)$$

$$\frac{q}{m} = \frac{ZE}{DlB^2}$$

In 1897, J. J. Thomson used this method to determine the charge-to-mass ratio of the electron. He obtained a value of approximately:

$$\frac{e}{m} \approx 1.76 \times 10^{11}\ \mathrm{C/kg}$$

The currently accepted value is:

$$\frac{e}{m} \approx 1.758820 \times 10^{11}\ \mathrm{C/kg}$$

The electric charge of the electron was later measured by Robert A. Millikan in his oil-drop experiment in 1909. Combining Millikan's measurement of the electron charge with Thomson's value of e/m allowed the mass of the electron to be determined.

The accepted values are:

- Mass of the electron: $m_e = 9.11 \times 10^{-31}\mathrm{kg}$
- Charge of the electron: $e = -1.60 \times 10^{-19}\ \mathrm{C}$

At present, there is no experimental evidence that the electron has an internal structure or is composed of smaller particles. Therefore, the electron is still considered a fundamental particle in the Standard Model of particle physics.

2. Nucleus

2.1. Discovery of the Nucleus

The study of nuclear physics began in the early twentieth century. In 1909, Hans Geiger and his student Ernest Marsden conducted an experiment under the supervision of Ernest Rutherford to investigate the scattering of alpha particles by a thin gold foil.

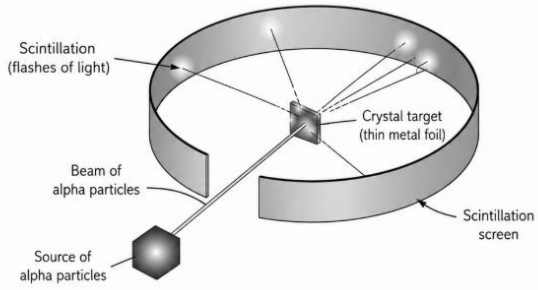


Figure 4.34 *Rutherford's alpha-particle scattering experiment*

A radioactive source of radium (Ra) was used to emit alpha particles toward a very thin gold foil. The scattered alpha particles were detected by the flashes of light produced on a zinc sulfide (ZnS) screen surrounding the gold foil. Geiger and Marsden carefully counted the flashes of light observed at different angles around the screen.

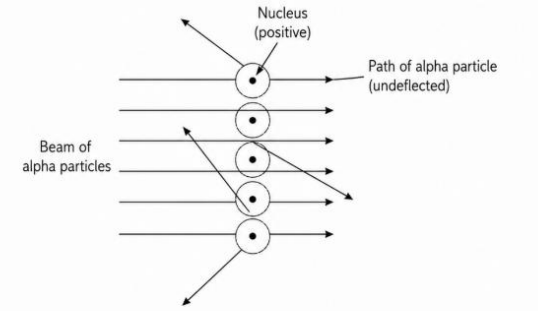


Figure 4.35 *Scattering of alpha particles by nuclei in a thin metal foil*

Most of the alpha particles passed straight through the gold foil without being deflected. Some alpha particles were deflected through small angles, while a very small number (about 1 in $8{,}000$) were scattered backward. Since alpha particles carry a positive charge of $+2e$ and have a mass about $8{,}000$ times greater than that of an electron, this result was very surprising.

Rutherford proposed that the large deflection and backward scattering of alpha particles could not be caused by the electrons because electrons have a very small mass compared with alpha particles. He concluded that the alpha particles must have collided with a small, dense, and positively charged region at the center of the atom.

From this experiment, Rutherford concluded that the atom contains a very small, dense nucleus with a positive charge. Although the nucleus occupies only a tiny fraction of the atom's volume, it contains almost all of the atom's mass.

Conclusion

The large-angle scattering and backward deflection of alpha particles provide evidence for the existence of a small, dense, and positively charged nucleus at the center of the atom.

In 1911, Ernest Rutherford proposed the nuclear model of the atom. He concluded that the atom consists of a small, heavy, positively charged nucleus surrounded by electrons moving around it. This discovery provided the first experimental evidence for the existence of the atomic nucleus and was made about 14 years after J. J. Thomson discovered the electron in 1897.

2.2. Constituent Particles of the Nucleus

The atomic nucleus is composed of two types of particles: protons and neutrons. These particles are collectively called nucleons.

2.2.1. Proton

The number of protons in the nucleus is called the atomic number (or proton number), denoted by Z. A proton is a positively charged particle. Its charge is equal in magnitude to the elementary charge.

Charge of a proton: $q_p = +e = +1.60 \times 10^{-19}\ \text{C}$

Mass of a proton: $m_p = 1.67 \times 10^{-27}\ \text{kg}$

The mass of a proton is approximately 1,836 times greater than the mass of an electron. The proton was identified by Ernest Rutherford in 1917.

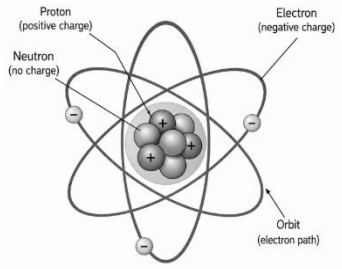


Figure 4.36 *Structure of an atom*

2.2.2. Neutron

The neutron was discovered by James Chadwick in 1932. The number of neutrons in the nucleus is denoted by N. A neutron is an electrically neutral particle, which means it has no electric charge. Its mass is approximately equal to the mass of a proton.

Charge of a neutron: $q_n = 0$

Mass of a neutron: $m_n = 1.675 \times 10^{-27}\ \text{kg}$

The neutron has no electric charge and is slightly more massive than the proton. Together with protons, neutrons make up the nucleus of an atom.

3. Photoelectric Effect

The development of quantum theory was greatly influenced by the contributions of Albert Einstein and Louis de Broglie. At the beginning of the twentieth century, their work introduced a new understanding of the dual nature of light and matter, leading to the development of wave mechanics, which became an important part of quantum mechanics.

3.1. Illustration of the Photoelectric Effect

The photoelectric effect was first observed by Heinrich Hertz in 1887. This phenomenon provided important evidence that light can behave as a stream of particles called photons.

3.1.1. Experiment

An electroscope is positively charged, and a zinc plate is attached to its metal cap.

Ultraviolet light from a light source, such as a mercury-vapour lamp, is directed onto the zinc plate. It is observed that the electroscope discharges rapidly, causing the gold leaves of the electroscope to collapse.

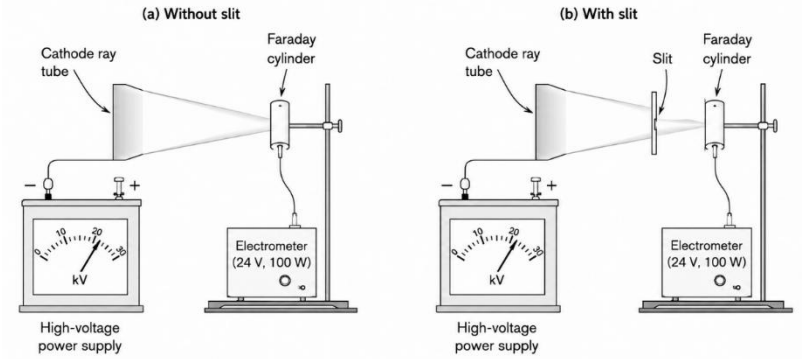


Figure 4.37 *Measurement of the charge carried by cathode rays*
The Faraday cylinder collects the charge. With a slit, a smaller current is collected.

When an ordinary glass sheet is placed between the light source and the zinc plate, the electroscope no longer discharges, and the gold leaves remain separated. This is because ordinary glass absorbs most ultraviolet radiation, preventing the ultraviolet light from reaching the zinc plate.

3.1.2. Explanation

• When the zinc plate and the metal cap of the electroscope are negatively charged, they contain an excess of electrons. When ultraviolet light shines on the zinc plate, electrons are emitted from its surface. As a result, the electroscope gradually loses its negative charge. This phenomenon is called the photoelectric effect.

• When a glass sheet is placed between the light source and the zinc plate, the glass absorbs most of the ultraviolet radiation. As a result, electrons cannot be emitted from the zinc plate, and the photoelectric effect does not occur. This experiment shows that the photoelectric effect depends on the wavelength (or frequency) of the incident light. It occurs only when the light has a sufficiently short wavelength (or high frequency), which means the photons have enough energy to eject electrons from the metal surface.

3.2. Einstein's Explanation of the Photoelectric Effect

Albert Einstein explained the photoelectric effect by proposing that light consists of particles called photons. Each photon carries a discrete amount of energy, called a quantum of energy, and travels at the speed of light in a vacuum.

To eject an electron from the surface of a metal, the energy of the incident photon, hf, must be greater than or equal to the work function E_0 of the metal. Therefore,

$$hf \geq E_0$$

where:

- $h = 6.626 \times 10^{-34}$ J.s is Planck's constant,
- f is the frequency of the incident light, measured in hertz (Hz).
- E_0 is the work function of the metal, measured in joules (J).

This equation shows that the frequency of the incident light must be high enough for the photoelectric effect to occur. If the energy of the photon is less than the work function of the metal ($hf < E_0$) no electrons will be emitted regardless of the intensity of the light.

This explanation is different from the classical wave theory of light, which predicted that increasing the intensity of light should eventually provide enough energy to eject electrons at any frequency. However, experiments show that the photoelectric effect occurs only when the frequency of the incident light is greater than or equal to the threshold frequency of the metal.

If the photon energy is greater than the work function, the remaining energy appears as the maximum kinetic energy of the emitted electron, given by Einstein's photoelectric equation:

$$K_{\max} = \frac{1}{2}mv^2 = hf - E_0$$

where:

- $K_{\max}$ is the **maximum kinetic energy** of the emitted electron.
- m is the **mass of the electron**.
- v is the **maximum speed** of the emitted electron.

4. Bohr Atomic Model

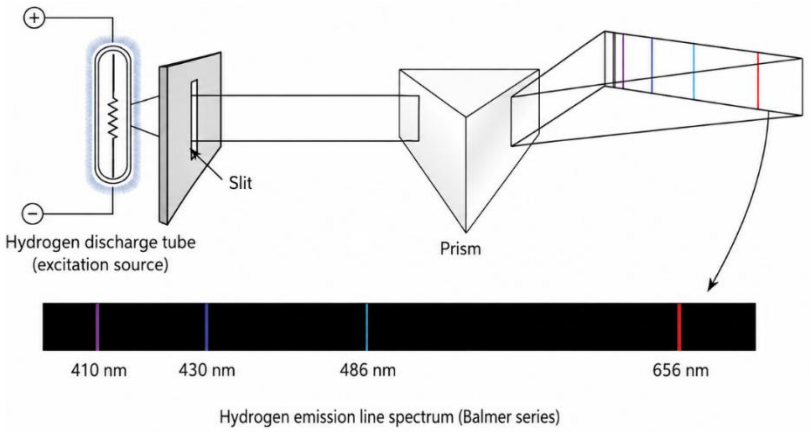


Figure 4.38 *Production of the hydrogen line spectrum using a prism*

The **hydrogen emission spectrum** was first observed in **1853**. However, it was not fully explained until the beginning of the twentieth century, when **Max Planck** and **Niels Bohr** introduced new ideas about atomic structure and quantum theory.

Based on **Rutherford's nuclear model**, Bohr proposed that a hydrogen atom consists of a single electron revolving around a small, positively charged nucleus.

According to Rutherford's model, the electron was compared to a satellite orbiting the Earth. If a satellite continuously loses energy while orbiting, its orbit gradually decreases until it eventually falls to Earth. Similarly, according to classical physics, an electron revolving around the nucleus should continuously radiate energy, causing its orbit to shrink until it finally falls into the nucleus. However, this does not happen in reality, indicating that Rutherford's model could not completely explain the stability of atoms.

Bohr proposed that **classical physics cannot completely describe the behavior of electrons inside an atom**. He suggested that electrons move only in certain permitted circular orbits, called

stationary orbits or **energy levels**. While an electron remains in one of these permitted orbits, it does **not** emit or lose energy.

Bohr further proposed that an electron can absorb or emit energy only when it jumps from one permitted orbit to another. The energy is absorbed or emitted in the form of electromagnetic radiation (a photon). Therefore, the **line spectrum** of an element is produced when electrons transition between different energy levels.

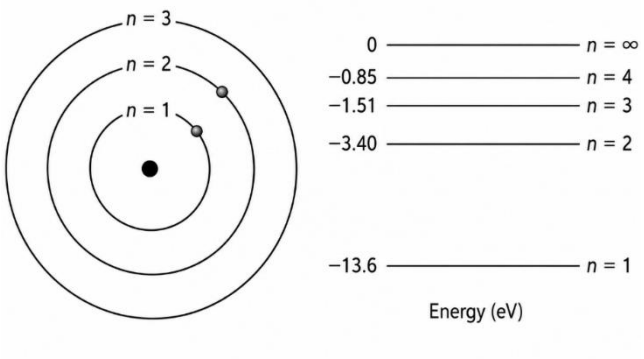


Figure 4.39 *Bohr model of the hydrogen atom and the corresponding energy levels*

Each permitted orbit corresponds to a specific energy value. The energy of each orbit is negative because the electron is bound to the nucleus by the electrostatic force. Energy must be supplied to remove the electron completely from the atom. The orbit closest to the nucleus has the **most negative energy**, meaning the electron is most tightly bound and requires the greatest amount of energy to remove it from the atom. An electron that has completely escaped from the atom has zero energy.

Since each orbit has a definite energy, an electron cannot exist between two permitted energy levels. Bohr proposed that electrons can gain or lose energy only in discrete amounts, called **quanta**. This idea became one of the fundamental principles in the development of **quantum mechanics**.

In a hydrogen atom, only one electron revolves around the nucleus. When the electron occupies the lowest energy level $(n = 1)$, the atom is said to be in its **ground state** (or fundamental state). At this energy level, the electron cannot lose any more energy, making it the most stable state of the hydrogen atom.

Example

The ground state $(n = 1)$ is the lowest energy level of the hydrogen atom. An electron in this level requires **13.6 eV** of energy to be completely removed from the atom. This minimum energy is called the **ionization energy** of hydrogen.

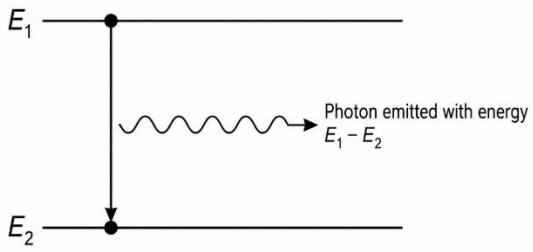


Figure 4.40 *Emission of a photon during a transition between energy levels*

An electron can move to a higher energy level by absorbing energy. This process is called **excitation**, and the atom is said to be in an **excited state**. The required energy may be supplied through collisions with other particles, such as free electrons, or by absorbing electromagnetic radiation.

An electron in an excited state is unstable. It eventually returns to a lower energy level by emitting a photon. The energy of the emitted photon is equal to the difference between the two energy levels.

If the electron absorbs energy equal to or greater than the ionization energy, it completely escapes from the atom. This process is called **ionization**.

Bohr's atomic model explains that electrons move from one orbit to another only by absorbing or emitting a specific amount of energy. Since electrons cannot exist between two permitted energy levels, only photons with specific energies (or frequencies) can be emitted or absorbed. This explains why the hydrogen atom produces a **line emission spectrum** instead of a continuous spectrum.

Bohr also predicted that the energy levels become closer together as the principal quantum number $n$increases and approach zero as n approaches infinity. His atomic model successfully explained the hydrogen emission spectrum and laid the foundation for the development of modern quantum mechanics.